Delhivery Data Processing & Feature Engineering Case Study

Dataset Overview

Shape: (17786, 24)

trip_creation_time ... segment_factor count 17786 ... 17786.000000 mean 2026-03-23 11:24:06.814348288 ... 2.188171 min 2026-03-23 00:00:18 ... -4.933333 25% 2026-03-23 05:10:24 ... 1.333333 50% 2026-03-23 11:04:36 ... 1.666667 75% 2026-03-23 17:37:00 ... 2.210526 max 2026-03-23 23:59:54 ... 192.500000 std NaN ... 4.414769 [8 rows x 14 columns]

Data Cleaning

Removed duplicates, handled missing values, and converted timestamps.

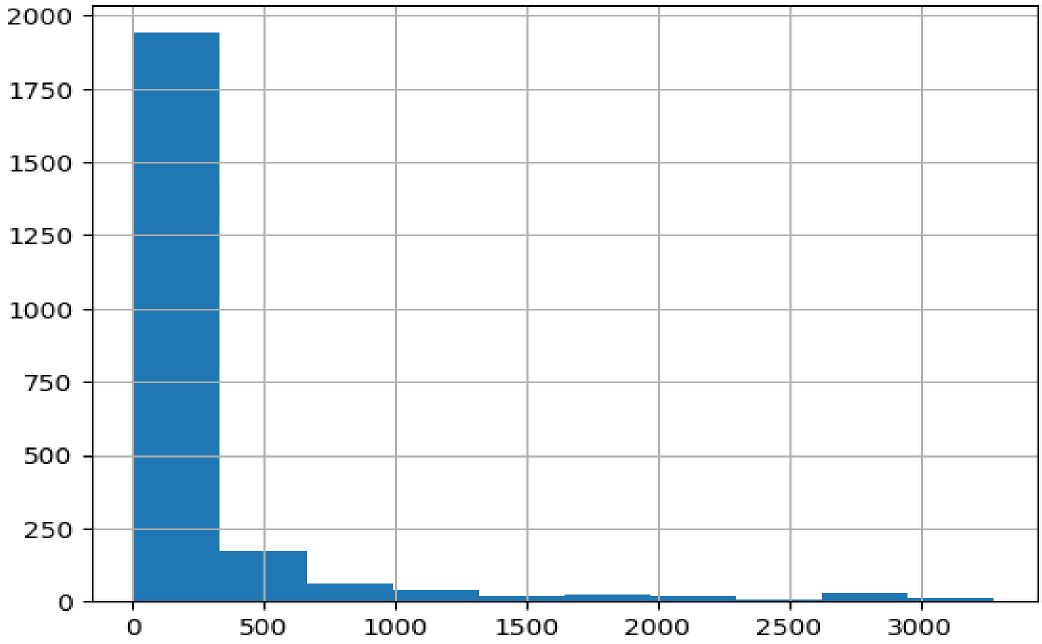
Aggregation

Trips aggregated across segments to compute total time and distance.

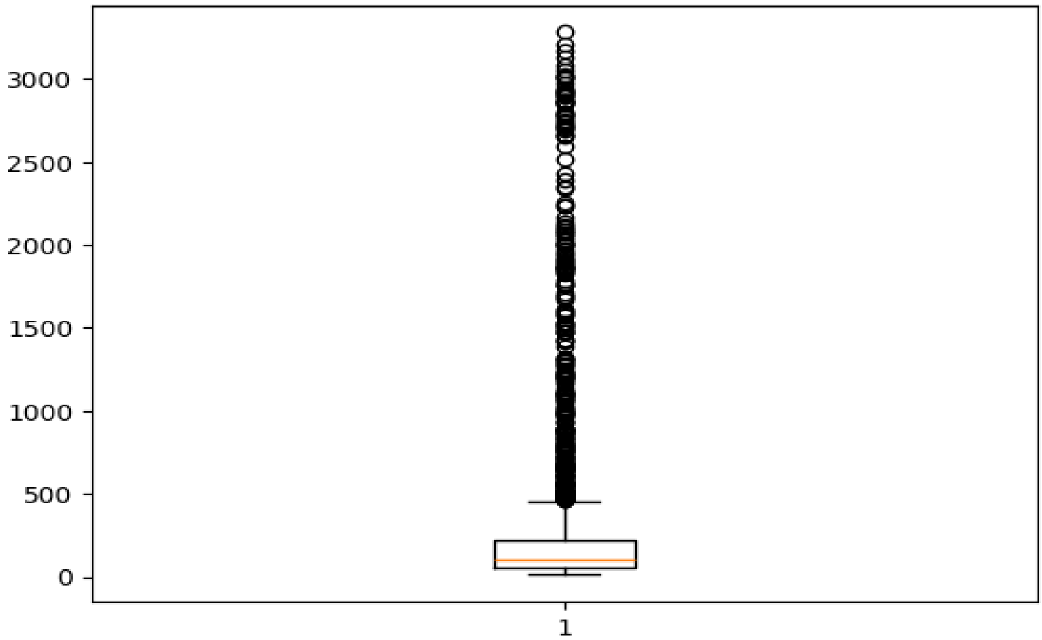
Feature Engineering

Extracted time-based features and computed trip duration.

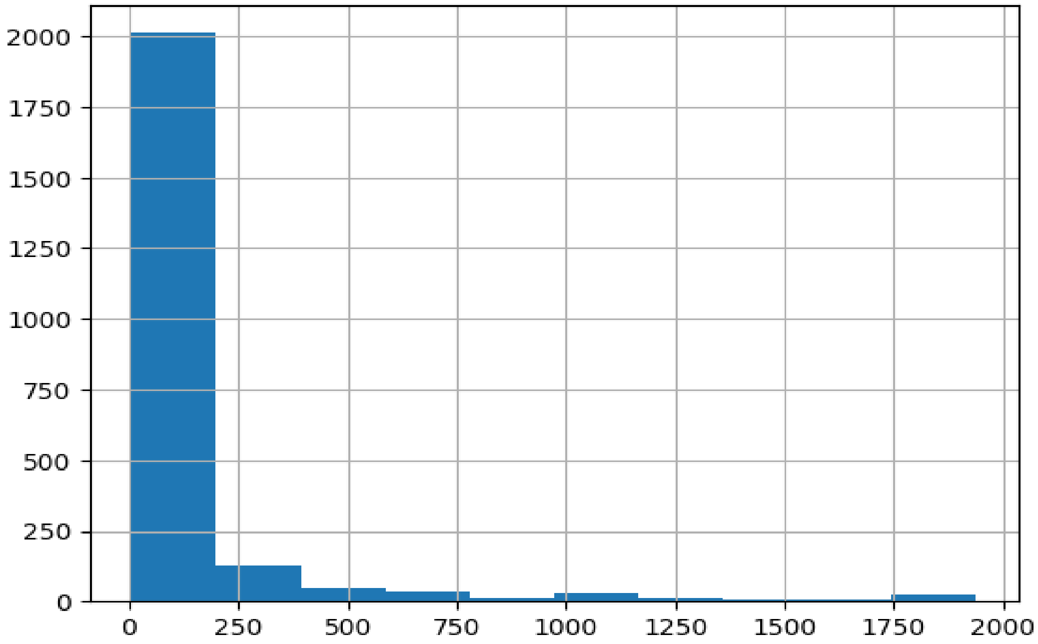
Univariate Analysis



Distribution shows skewness and spread.



Boxplot reveals outliers.



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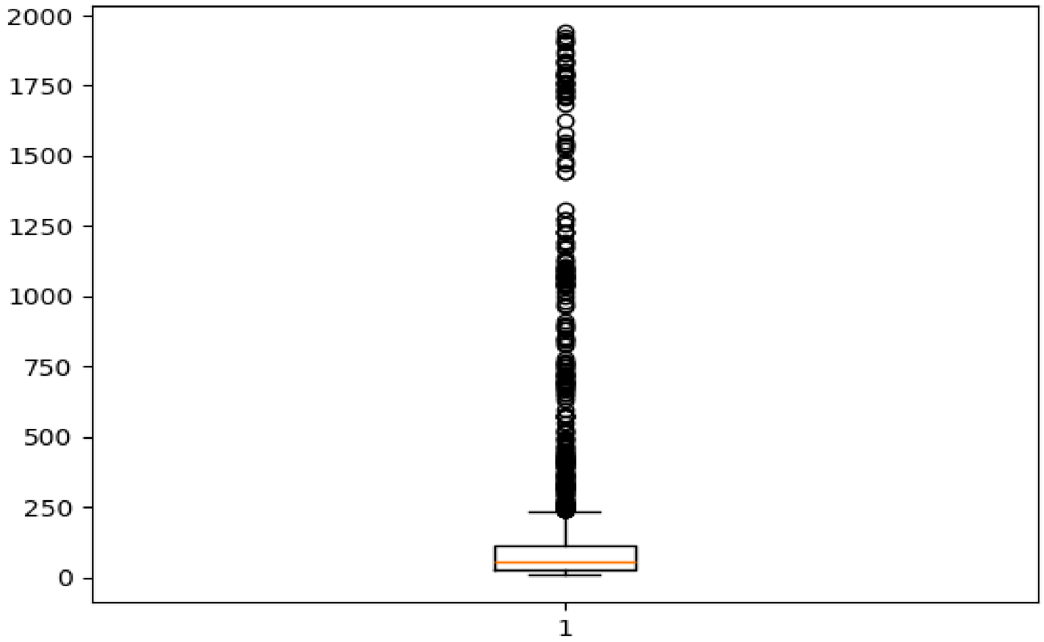

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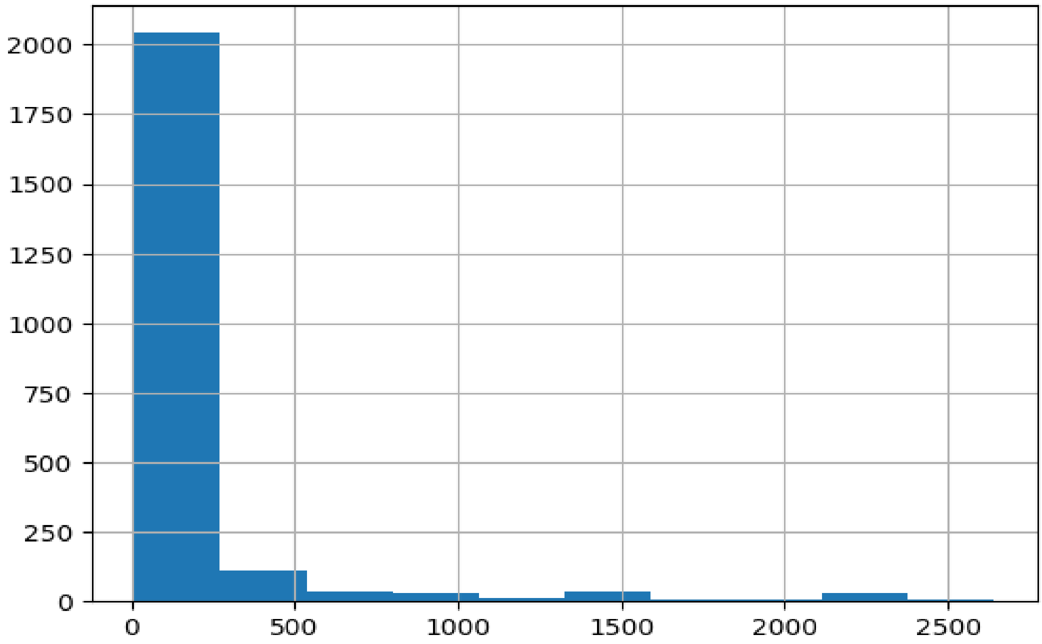

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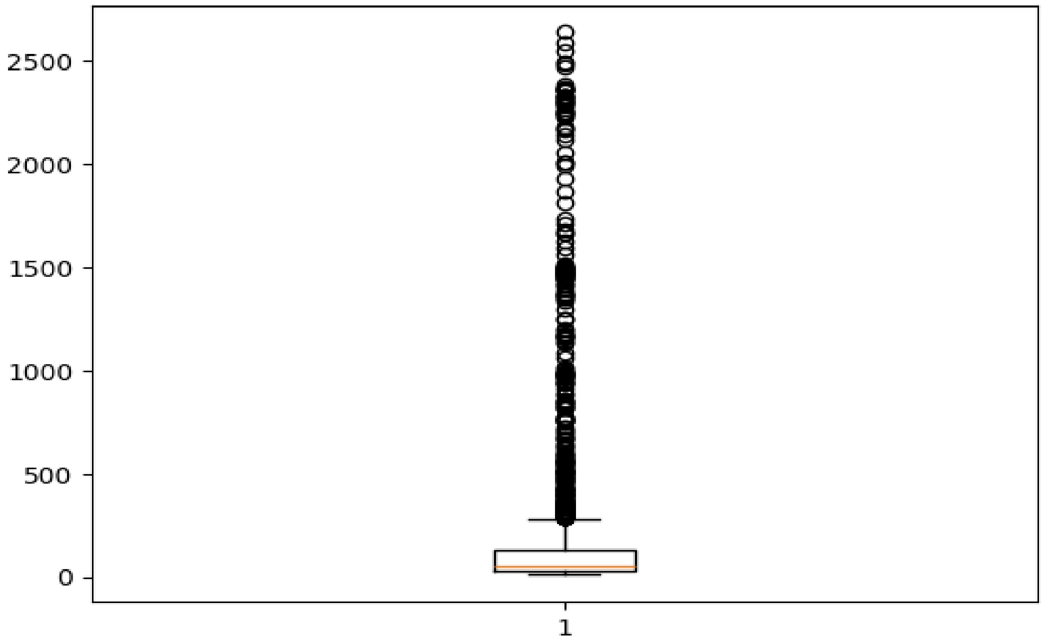
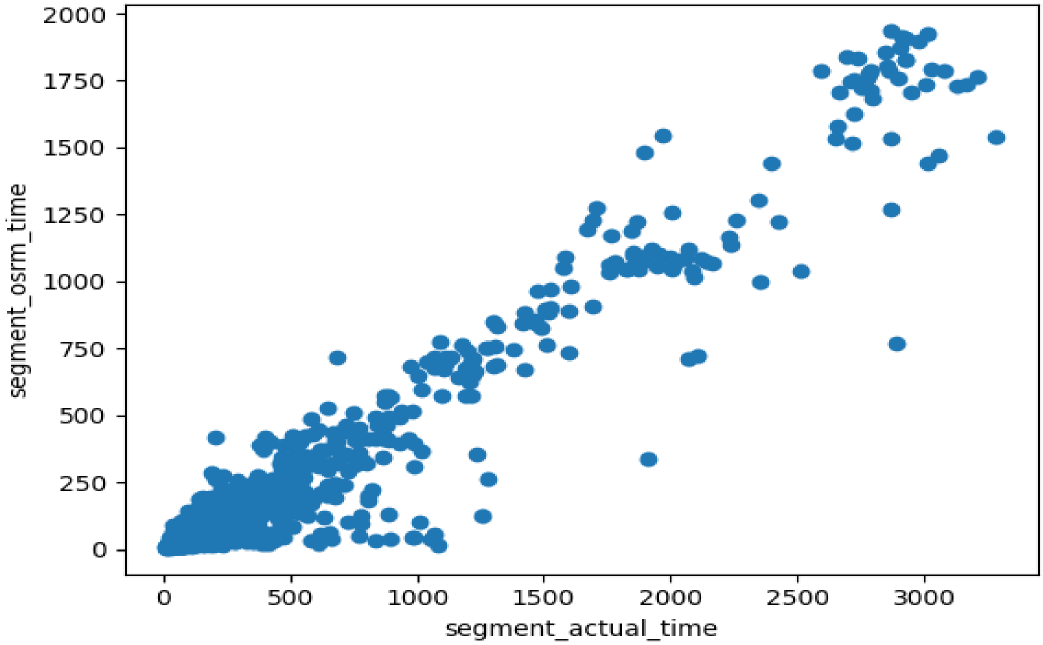
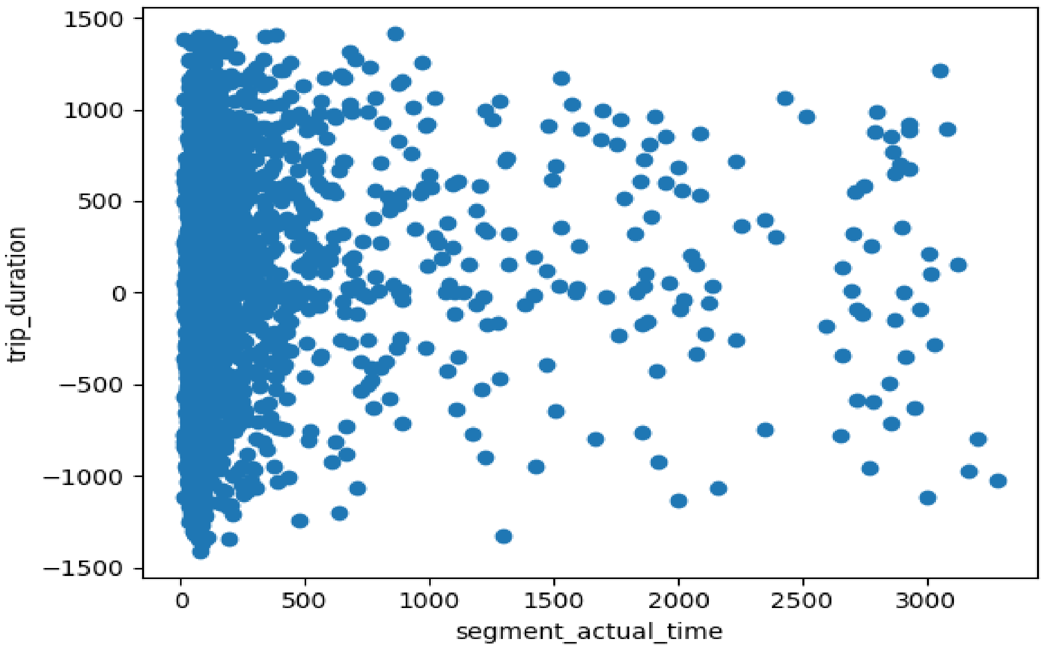


Boxplot reveals outliers.

Relationship Analysis



Scatter plot shows relationship trend.



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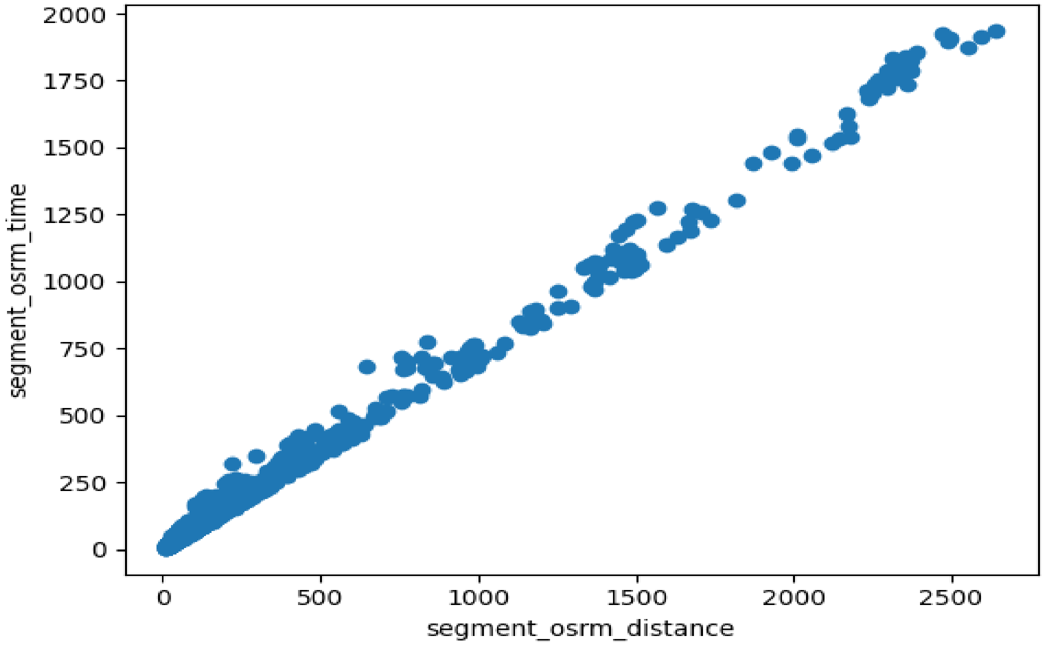

Scatter plot shows relationship trend.

Hypothesis Testing

T-test actual vs osrm: t=26.69, p=0.00000

T-test actual vs duration: t=10.58, p=0.00000

Insights

Differences exist between actual and estimated metrics indicating inefficiencies.

Business Recommendations

Improve routing algorithms and prediction systems.